

Muhammad Anas Khan

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EDUCATION

BS Computer Science — FAST-NUCES, Karachi

2022 – 2026 | CGPA: 3.13

EXPERIENCE

Associate DevOps Engineer — QBS Co., Karachi

Dec 2025 – Present

- Build and maintain CI/CD pipelines, containerized deployments (Docker/Kubernetes), and cloud-native and on-prem server infrastructure.
- Develop gRPC APIs, integrate MLOps workflows, and maintain Kubernetes clusters with DevSecOps practices to ensure secure, scalable backend systems.
- Monitor distributed services via Grafana/Loki/Prometheus and build internal infrastructure tooling to surface operational insights.

Web Development Intern (MERN Stack) — OutSecure, Rawalpindi (Remote)

Jul – Sep 2025

- Built React and MERN stack features in a collaborative, industry-standard workflow during a structured remote internship.

SKILLS

Languages: Python, C++, JavaScript, Go, Bash, Kotlin

Web & Backend: React, MERN Stack, FastAPI, REST, gRPC, Microservices

Databases & Cloud: PostgreSQL, MongoDB, MySQL, AWS (VPC, EC2, S3, ASG, IAM)

DevOps: Docker, Kubernetes, Jenkins, ArgoCD, Terraform, Grafana, Loki, Prometheus, OTel, Linux

ML / AI: NumPy, Pandas, Scikit-learn, PyTorch, Keras, TensorFlow

PROJECTS

Offline Payment System for Pakistan | Final Year Project

2026

Tech: FastAPI · PostgreSQL · BLE · Kotlin · Jetpack Compose · Android Keystore / TEE | [GitHub \(App\)](#) [GitHub \(Backend\)](#)

- FastAPI backend with nonce + BLE double-spending prevention via asymmetric crypto and hash-chained ledgers; Kotlin/Jetpack Compose Android app for P2P payments over QR + BLE, secured by Android Keystore and TEE.

Production Infrastructure & Tooling | Proprietary — QBS Co.

2026

Tech: Go · gRPC · OpenTelemetry · Jaeger · Kubernetes · Grafana · Loki · Fluent Bit · Jenkins · Trivy

- gRPC log collection service in Go with worker pool, panic isolation, and OTel/Jaeger distributed tracing across a 4-service chain.
- On-prem Kubernetes logging stack (Grafana, Loki, Fluent Bit) with LogQL dashboards; Jenkins CI/CD pipeline with Trivy security scanning and Slack notifications.

Visual Daily Planner | Accessibility Web App

2025

Tech: React · Redux · Firebase / Firestore · Tailwind CSS · Shadcn UI | [Live Demo](#) [GitHub](#)

- Full-stack routine planner for autistic youth with Firebase Auth, Firestore real-time sync, midnight auto-reset with history logging, and Redux state management; deployed on Vercel.

Bacterial Colony Classifier | Deep Learning

2025

Tech: Python · TensorFlow · Keras · EfficientNetB0 · Streamlit | [GitHub](#)

- EfficientNetB0 transfer learning model classifying 33 bacterial colony types from a class-imbalanced dataset; data augmentation and CNN fine-tuning with a Streamlit demo.

Multi-Store Inventory Management System | Backend

2025

Tech: Node.js · Express.js · PostgreSQL · Redis · JWT | [GitHub](#)

- RESTful backend for 500–1,000+ stores with JWT auth, Redis caching, async audit logging, rate limiting, bcrypt hashing, and SQL injection-safe queries.

CERTIFICATIONS & ACHIEVEMENTS

- KodeKloud — Docker for Absolute Beginners & Kubernetes for Absolute Beginners

2025

- Dean's List of Honours — FAST-NUCES Karachi, Semester 7 (CGPA 3.67 / 4.0)

2025

- Qualified Round 2 — Speed Programming, NUCES Coder Cup (~40–50 teams, solved all 5 problems within cutoff)

2023

- Top 4 Identifier & Top 13 Observer from Pakistan — iNaturalist (volunteer naturalist)

Ongoing